

Classwork (11th August)

1. A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles. A marble is chosen at random from the jar. After replacing it, a second marble is chosen. What is the probability of choosing a green and then a yellow marble?
2. A fair coin is tossed two times. Construct the sample space of the experiment. What is the probability of getting (i) all head (ii) at least one head (iii) at best one head (iv) a head and a tail.
3. Two dice are thrown at random. Find the probability that:
 - (i) The total of the numbers on the dice is 8
 - (ii) A doublet (two dice landing on the same value)
 - (iii) A sum greater than 5
 - (iv) A sum less than 4 or greater than 8
 - (v) An even number on the first die
 - (vi) The difference between the numbers of two dice is 3
 - (vii) At least one 6
 - (viii) At least one 6, if the two faces are different
4. Tickets are numbered from 1 to 100. They are well shuffled, and a ticket is drawn at random. What is the probability that the drawn ticket has
 - (i) An even number
 - (ii) A number 5 or a multiple of 5
 - (iii) A number which is greater than 75
 - (iv) A number which is a square
5. In a certain town, out of 10000 people residing, 2400 are above 25 years of age and 6000 are males. Out of the 2400 who are above 25 years of age, 400 are males. Suppose after a person is chosen you are told that the person is male. What is the probability that he is above 25 years of age?
6. A jar consists of 21 sweets. 12 are green and 9 are blue. William picked two sweets at random.
 - (i) Draw a tree diagram to represent the experiment.
 - (ii) Find the probability that
 - a) both sweets are blue.
 - b) One sweet is blue, and one sweet is green.
 - (iii) William randomly took a third sweet. Find the probability that:
 - a) All three sweets are green.
 - b) At least one of the sweets is blue.
7. A box contains seven balls – two red (R), three blue (B) and two yellow (Y). Consider an experiment that consists of drawing a ball from the box.
 - (i) What is the probability that the first ball drawn is yellow?
 - (ii) What is the probability that the same-coloured ball is drawn twice with replacement?
 - (iii) What is the probability that the same-coloured ball is drawn twice without replacement?

8. Suppose a study of speeding violations (last year) and drivers who use cell phones produced the following fictional data:

	Speeding violation	No speeding violation	Total
Cell phone user	25	280	305
Not a cell phone user	45	405	450
Total	70	685	755

- (a) Find the probability that a randomly selected person is a cell phone user.
- (b) What is the probability that a randomly selected person had no speeding violation last year?
- (c) If a randomly selected person had a speeding violation last year, what is the probability that he is a cell phone user?
- (d) If a randomly selected person is not a cell phone user, what is the probability that he had a speeding violation last year?
- (e) If a randomly selected person had a speeding violation last year, what is the probability that he is not a cell phone user?
- (f) Are the events "Cell phone user" and "Speeding violation" mutually exclusive? Why or why not?
- (g) Determine if the events "Not a cell phone user" and "No speeding violation" are independent.
9. During the monsoon season in a coastal city, there is a 60% chance of rain on any given day. When it rains, there is a 50% chance that strong winds will accompany the rain. On days when it does not rain, there is still a 20% chance that the city will experience strong winds due to nearby storms.
- (a) What is the probability that it will not rain on a random day during the monsoon season?
- (b) What is the probability that strong winds will not occur when it does not rain?
- (c) What is the probability that, on a random day of the monsoon season, it will rain and strong wind?
- (d) What is the probability that it will not rain and there will be no strong winds on a random day?
- (e) Suppose there is a 38% chance that there will be strong winds on a random day regardless of whether it rains or not. Determine the probability that it rained that day if strong winds are observed.
10. A certain delivery service offers both express and standard delivery. Seventy-five percent of parcels are sent by standard delivery, and 25% are sent by express. Of those sent standard, 80% arrive the next day, and of those sent express, 95% arrive

the next day. A record of a parcel delivery is chosen at random from the company's files.

- (a) What is the probability that the parcel was shipped express and arrived the next day?
- (b) What is the probability that it will arrive the next day?
- (c) Given that the package arrived the next day, what is the probability that it was sent express?

11. A certain level of customer dissatisfaction is present in about 1 out of 1000 customers in a given population. Suppose there is a customer feedback survey, and customers are asked to rate their satisfaction level. The survey results show that a positive rating is given with a probability of 10% for a dissatisfied customer and with a probability of 95% for a satisfied customer.

- (a) If a customer is selected at random from this population, what is the probability that the feedback survey will result in a positive rating? Interpret your findings.
- (b) If the feedback survey of a selected customer results in a negative rating, what is the probability that they are dissatisfied?

12. Two factories supply light bulbs to the market. Bulbs from factory X work for over 5000 hours in 99% of cases, whereas bulbs from factory Y work for over 5000 hours in 95% of cases. It is known that factory X supplies 60% of the total bulbs available in the market.

- (a) What is the probability that a purchased bulb will work for longer than 5000 hours?
- (b) Given that a light bulb works for more than 5000 hours, what is the probability that it was supplied by factory Y?
- (c) Given that a light bulb work does not work for more than 5000 hours, what is the probability that it was supplied by factory X?